

Yao Meng

Senior Software Development Engineer - Distributed Storage System

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Summary

Senior Software Engineer specializing in distributed storage systems and high-performance caching systems.

Founder and maintainer of **Foyer** (1.6k ★), hybrid cache in Rust, used in **Apache OpenDAL**, **RisingWave**, **ChromaDB**, **SlateDB**, **ZeroFS**, etc.

Active open-source contributor, contributed to **Apache OpenDAL**, **TiKV**, **Chroma**, **RisingWave**, **SlateDB**, etc.

Education

Beihang University

M.S. in Computer Science and Engineering

Beijing, China

Sep. 2020 - Jan. 2023

- GPA: 3.80/4.0

Beihang University

B.S. in Computer Science and Engineering

Beijing, China

Sep. 2016 - Jun. 2020

- GPA: 3.72/4.0, Rank 6/157 (Top 4%); Outstanding Graduate (2 in 157).

Project

Foyer - Hybrid Cache Library in Rust

<https://github.com/foyer-rs/foyer>

Founder and Maintainer

May. 2023 - now

- Designed and built a production-ready hybrid (memory + disk) cache library in Rust, targeting cloud-native data infrastructure.
- Achieved high performance via a plug-and-play modular architecture, replaceable algorithms and engines, and optimized concurrency.
- Built with enterprise features such as rich observability, extensive configuration options, and broad ecosystem compatibility.
- 1.6k ★, used by 100+ projects on Github, including **Apache OpenDAL**, **RisingWave**, **ChromaDB**, **SlateDB**, **ZeroFS**, **Percas**, etc.
- A deep dive into Foyer: <https://blog.mrcroxx.com/posts/foyer-a-hybrid-cache-in-rust-past-present-and-future/>

Experience

RisingWave Labs

Beijing, China

Database Kernel Development Engineer - Rust

Nov. 2021 - Mar. 2026

- Early engineer at RisingWave, contributing from the ground up to the design and evolution of the core storage and streaming system.
- Owned the design and implementation of the hybrid cache subsystem in RisingWave's storage engine.
- Led optimization of S3-based LSM-tree storage engine read path, achieve 3x more throughput for join-heavy workload, reducing S3 read operations by 90% and overall storage cost by 45%, and optimized system stability.

Chroma

Remote

Part-time Consultation

- Provided design guidance on the hybrid cache system, integrated Foyer into Chroma, involved in debugging and technical support.

PingCAP

Beijing, China

Distributed Storage System Development Intern - Rust

Apr. 2021 - Nov. 2021

- Optimized the write-ahead log (WAL) recovery path in TiDB/TiKV, achieving up to 10× faster recovery time.
- Improved algorithms in the Raft-based key-value storage engine, reducing leader eviction latency by 2× under slow-follower scenarios.
- Optimized the Raft log GC of the RocksDB-based WAL engine, reducing Raft log GC bandwidth usage by 5%–30% across different workloads.
- Contributed to the maintenance of PingCAP's RocksDB fork and the Rust bindings used in TiKV.

Bytedance

Beijing, China

Backend Development Intern - Go

Oct. 2019 - Jan. 2020

- Contributed to backend services and data infrastructure for a large-scale advertising recommendation system.

Presentation

Apache Community Over Code Asia 2025

Beijing

Enhancing Performance and Reducing Cost for Object Store Access with Apache OpenDAL and Foyer

Jul. 27, 2025

Rust China Conf 2024

Shanghai

Foyer: Easily Enhance Your Cloud Data Engine with foyer Hybrid Cache

Sep. 07, 2024